Chapter 8: Duality

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(This slide is mainly modified from slides of S. Boyd EE364a, Stanford, and R. Tibshirani 10-725, CMU.)

Reap: duality in linear programs

Given $c \in \mathbb{R}^n, A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$:

Primal LP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b \\ & x \succeq 0 \end{array}$$

Dual LP

$$\begin{array}{ll} \text{maximize} & -b^T \nu \\ \text{subject to} & A^T \nu - \lambda + c = 0 \\ & \lambda \succeq 0 \end{array}$$

Lagrangian

Consider the general optimization problem (not necessarily convex)

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_j(x) = 0, \quad j = 1, \ldots, p \end{array}$$

where variable $x \in \mathbb{R}^n$, domain $\mathcal{D}$, optimal value p^*.

Lagrangian: $L : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R}$,

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{j=1}^{p} \nu_j h_j(x)$$

where λ_i, ν_j are called Lagrange multipliers.

Lagrange dual function

Lagrange dual function: $g : \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R}$

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) = \inf_{x \in \mathcal{D}} \left(f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{j=1}^{p} \nu_j h_j(x) \right)$$

g is concave, which can be $-\infty$ for some λ, ν.

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g is concave, which can be $-\infty$ for some λ, ν.

lower bound property: if $\lambda \succeq 0$, then $g(\lambda, \nu) \leq p^*$.
proof: if $\tilde{x}$ is feasible and $\lambda \succeq 0$, then

$$f_0(\tilde{x}) \geq L(\tilde{x}, \lambda, \nu) \geq \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) = g(\lambda, \nu)$$

minimizing over all feasible $\tilde{x}$ gives $p^* \geq g(\lambda, \nu)$.

Example: standard form LP

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b, \quad x \succeq 0 \end{array}$$

dual function

- Lagrangian is

$$L(x, \lambda, \nu) = c^T x + \nu^T (Ax - b) - \lambda^T x = -b^T \nu + (c + A^T \nu - \lambda)^T x$$

- L is affine in x, hence

$$g(\lambda, \nu) = \inf_x L(x, \lambda, \nu) = \begin{cases} -b^T \nu & \text{if } A^T \nu - \lambda + c = 0 \\ -\infty & \text{otherwise} \end{cases}$$

  g is linear on affine domain $\{(\lambda, \nu) : A^T \nu - \lambda + c = 0\}$, hence concave.

lower bound property: $p^* \geq -b^T \nu$ if $A^T \nu + c \succeq 0$

Example: least-norm solution of linear equations

$$\begin{array}{ll} \text{minimize} & x^T x \\ \text{subject to} & Ax = b \end{array}$$

dual function

- Lagrangian: $L(x, \nu) = x^T x + \nu^T (Ax - b)$
- To minimize L over x, set gradient equal to zero:

$$\nabla_x L(x, \nu) = 2x + A^T \nu = 0 \;\Rightarrow\; x = -\frac{1}{2} A^T \nu$$

- Plugin into L to obtain g: $g(\nu) = -\frac{1}{4} \nu^T A A^T \nu - b^T \nu$, concave in ν

lower bound property: $p^* \geq -\frac{1}{4} \nu^T A A^T \nu - b^T \nu$ for all ν

The dual problem

Lagrange dual problem:

$$\begin{array}{ll} \text{maximize} & g(\lambda, \nu) \\ \text{subject to} & \lambda \succeq 0 \end{array}$$

- Its optimal value denoted by d^*, finds the best lower bound on p^*
- λ, ν are dual feasible if $\lambda \succeq 0$, $(\lambda, \nu) \in \text{dom} g$
- The dual problem is a convex optimization problem, since

$$g(u, v) = -\max_x \{-f(x) - \sum_{i=1}^{m} \lambda_i f_i(x) - \sum_{j=1}^{p} \nu_j h_j(x)\}$$

which is the pointwise maximum of convex functions in (λ, ν).

Weak and strong duality

Weak duality: $d^* \leq p^*$

- always holds (even if primal problem is nonconvex).

strong duality: $d^* = p^*$

- does not hold in general
- (usually) holds for convex problems
- guarantee conditions are called constraint qualifications

Slater's constraint qualification

Slater's condition: if the primal is convex (i.e. f_0, f_i are convex, h_j are affine), and there exists at least one strictly feasible $x \in \mathbb{R}^n$, i.e.

$$\exists x \in \mathrm{int}\mathcal{D}, \text{ s.t. } f_i(x) < 0, i = 1, \ldots, m, \quad Ax = b,$$

the strong duality holds.

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the strong duality holds.

Refinement: strict inequalities only need to hold over functions f_i that are not affine.

Duality gap

Given primal feasible x and dual feasible λ, ν, the quantity $f_0(x) - g(\lambda, \nu)$ is called the duality gap between x and λ, ν.

Note that

$$f_0(x) - p^* \leq f_0(x) - g(\lambda, \nu)$$

so if the duality gap is zero, then x is primal optimal, and similarly, λ, ν are dual optimal.

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For an algorithmic viewpoint, it provides a stopping criterion:

if $f_0(x) - g(\lambda, \nu) \leq \epsilon$, then we are guaranteed that $f_0(x) - p^* \leq \epsilon$.

Complementary slackness

Assume strong duality holds, x^* is primal optimal, (λ^*, ν^*) is dual optimal, then

$$\begin{aligned} f_0(x^*) = g(\lambda^*, \nu^*) &= \inf_x \left(f_0(x) + \sum_{i=1}^m \lambda_i^* f_i(x) + \sum_{j=1}^p \nu_j^* h_j(x) \right) \\ &\leq f_0(x^*) + \sum_{i=1}^m \lambda_i^* f_i(x^*) + \sum_{j=1}^p \nu_j^* h_j(x^*) \\ &\leq f_0(x^*) \end{aligned}$$

hence, the two inequalities hold with equality

- x^* minimizes $L(x, \lambda^*, \nu^*)$;
- $\lambda_i^* f_i(x^*) = 0$ for $i = 1, \ldots, m$ (known as complementary slackness)

$$\lambda_i^* > 0 \Longrightarrow f_i(x^*) = 0, \qquad f_i(x^*) < 0 \Longrightarrow \lambda_i^* = 0$$

Karush-Kuhn-Tucker (KKT) conditions

The following four conditions are called the Karush-Kuhn-Tucker or KKT conditions (for a problem with differentiable f_i, h_j):

1. **primal feasibility**: $f_i(x) \leq 0, i = 1, \ldots, m$, $h_j(x) = 0, j = 1, \ldots, p$;
2. **dual feasibility**: $\lambda \succeq 0$;
3. **complementary slackness**: $\lambda_i f_i(x) = 0, i = 1, \ldots, m$;
4. **stationarity**: gradient of Lagrangian with respect to x vanishes

$$\nabla f_0(x) + \sum_{i=1}^{m} \lambda_i \nabla f_i(x) + \sum_{j=1}^{p} \nu_j \nabla h_j(x) = 0.$$

KKT condition: necessity

If x^* and λ^*, ν^* are primal and dual optimal, with zero duality gap, then x^*, λ^*, ν^* must satisfy the KKT conditions.

Since:

- x^* minimizes $L(x, \lambda^*, \nu^*)$;
- complementary slackness hold;
- primal and dual feasibility hold by virtue of optimality.

KKT conditions: sufficiency

If there exists x^*, λ^*, ν^* that satisfy the KKT conditions, then

$$\begin{aligned} g(\lambda^*, \nu^*) &= f_0(x^*) + \textstyle\sum_{i=1}^m \lambda_i^* f_i(x^*) + \sum_{j=1}^p \nu_j^* h_j(x^*) \\ &= f(x^*) \end{aligned}$$

where the first equality holds from stationarity, and the second holds from complementary slackness.

Hence the duality gap is zero, so

> If x^* and λ^*, ν^* satisfy the KKT conditions, then x^* and λ^*, ν^* are primal and dual optimal.

Example: quadratic with equality constraints

Consider $P \succeq 0$,

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}x^T P x + q^T x \\ \text{subject to} & Ax = 0 \end{array}$$

Solution: Convex problem, no inequality constraints, so by KKT conditions:

x is a solution if and only if

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x \\ \nu \end{bmatrix} = \begin{bmatrix} -q \\ 0 \end{bmatrix}$$

for some ν.

Thank you!



References and further reading

- S. Boyd and L. Vandenberghe (2004), “Convex optimization”, Chapter 5.
- Dimitri P. Bertsekas (2009). "Convex optimization theory", Chapters 4 and 5.